

Module: 13- Networking with Windows Server

25. Discuss the role of Windows Firewall in Windows Server and how to configure it.

Ans :- **Role:**

- **Windows Firewall** protects the server from unauthorized access.
- It filters incoming and outgoing network traffic based on rules.
- Helps prevent attacks like unauthorized logins, malware, and port scanning.
- Works with **Advanced Security** to provide IPsec (secure communication).

Configuration Steps:

1. Open **Control Panel → System and Security → Windows Defender Firewall**
2. Click **Advanced Settings**
3. Configure:
 - **Inbound Rules** (allow/block incoming traffic)
 - **Outbound Rules** (control outgoing traffic)
4. To allow a port:
 - Click **New Rule → Port**
 - Select TCP/UDP → Enter port number (e.g., 3389 for RDP)
 - Allow connection → Apply to profiles → Finish
5. Enable/Disable firewall:
 - Click **Turn Windows Firewall ON/OFF**

26. What is Network Address Translation (NAT) in Windows Server, and how do you configure it?

Ans:- **Concept:**

- **NAT (Network Address Translation)** converts private IP addresses into a public IP address.
- Allows multiple devices to share one internet connection.
- Improves security by hiding internal IP addresses.

Configuration Steps (Using RRAS):

1. Open **Server Manager**
2. Add Role → **Remote Access**
3. Install **Routing and Remote Access (RRAS)**
4. Open **RRAS Console**
5. Right-click server → **Configure and Enable RRAS**
6. Select **NAT**
7. Choose:
 - Public interface (internet)
 - Private interface (internal network)
8. Finish setup → Start service

27. Explain the concept of Dynamic Host Configuration Protocol (DHCP) and how to configure it in Windows Server 2016.

Ans:- **Concept:**

- DHCP automatically assigns IP addresses to devices.
- Reduces manual configuration errors.
- Provides IP, subnet mask, gateway, DNS automatically.

Configuration Steps:

1. Open **Server Manager**
2. Add Role → **DHCP Server**
3. Complete installation and **Authorize DHCP**
4. Open **DHCP Console**
5. Create a new scope:
 - Right-click IPv4 → **New Scope**
 - Enter IP range (e.g., 192.168.1.1 – 192.168.1.100)
 - Set subnet mask
 - Configure gateway and DNS
6. Activate scope

28. Describe the process of configuring DNS (Domain Name System) in Windows Server.

Ans:- **Concept:**

- DNS translates domain names (like google.com) into IP addresses.
- Essential for network communication.

Configuration Steps:

1. Open **Server Manager**
2. Add Role → **DNS Server**
3. Open **DNS Manager**
4. Create a new zone:
 - Right-click **Forward Lookup Zones** → **New Zone**
 - Choose **Primary Zone**
 - Enter domain name (e.g., example.com)
5. Add records:
 - **A Record** → maps hostname to IP
 - **CNAME** → alias
6. Configure client:
 - Set DNS IP as server IP

29. What is Server Manager, and how do you use it to manage servers in Windows Server?

Ans:- **Concept:**

- **Server Manager** is a central dashboard to manage Windows Server.
- Used for installing roles, monitoring, and configuring servers.

Uses:

- Add/Remove roles and features
- Monitor server health
- Manage multiple servers
- Configure services like DHCP, DNS, AD

How to Use:

1. Open **Server Manager**
2. Dashboard shows server status
3. Use:
 - **Manage → Add Roles and Features**
 - **Tools menu** for services (DNS, DHCP, etc.)
4. Add remote servers for centralized management

30. Discuss the role of Remote Desktop Services (RDS) in Windows Server 2016 or 2019 and how to configure it.

Ans:- **Role:**

- Allows users to remotely access a server or desktop.
- Useful for remote work and centralized application hosting.
- Supports multiple users accessing the same server.

Configuration Steps:

1. Open **Server Manager**
2. Add Role → **Remote Desktop Services**
3. Choose deployment:
 - **Quick Start** (simple setup)
 - **Standard Deployment** (advanced)
4. Install:
 - RD Session Host
 - RD Connection Broker
 - RD Web Access
5. Enable Remote Desktop:
 - Go to **System Settings → Remote Desktop → Enable**
6. Allow users:
 - Add users to **Remote Desktop Users group**

7. Connect using:

- Remote Desktop Connection (mstsc)